

Task 2 — Feature Engineering, Model Optimization & Performance Comparison

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1. Objective

This report documents the completion of Task 2 of the MainCrafts AI/ML Internship. The goal was to build an enhanced House Price Prediction system using the California Housing Dataset by applying feature engineering, training multiple regression models, and selecting the best-performing model using measurable evaluation metrics.

2. Dataset Overview

2.1 California Housing Dataset

The California Housing Dataset was used throughout this task. It contains real housing-related data collected from the 1990 California census.

Property	Details
Total Samples	20,640
Number of Features	8 input features
Target Variable	Median House Value (HousePrice)
Missing Values	None
Source	sklearn.datasets.fetch_california_housing

2.2 Input Features

- MedInc — Median income in block group
- HouseAge — Median house age in block group
- AveRooms — Average number of rooms per household
- AveBedrms — Average number of bedrooms per household
- Population — Block group population
- AveOccup — Average number of household members
- Latitude — Block group latitude
- Longitude — Block group longitude

3. Methodology

3.1 Feature Scaling

StandardScaler was applied to normalize all input features before training. This is a critical step because features in this dataset exist on very different numeric ranges (e.g., Population can be in thousands while AveRooms is typically 4–8). Without scaling, features with larger ranges would unfairly dominate the model's learning process.

- After scaling: all features have mean ≈ 0 and standard deviation ≈ 1
- Ensures fair contribution from every feature
- Required for Ridge Regression to work correctly
- Improves convergence speed and model stability

3.2 Train-Test Split

The dataset was split into 80% training (16,512 samples) and 20% test (4,128 samples) using random_state=42 for reproducibility. This ensures the model is evaluated on data it has never seen during training.

3.3 Models Trained

Model	Key Parameter	Purpose
Linear Regression	None (baseline)	Simple baseline model
Ridge Regression	alpha=1.0	L2 regularization to reduce overfitting
Decision Tree	max_depth=5	Captures non-linear relationships

4. Results & Performance Comparison

4.1 Evaluation Metrics Used

- RMSE (Root Mean Squared Error) — Measures average prediction error. Lower is better.
- R² Score — Measures how well the model explains variance in the target. Higher is better (max=1.0).

4.2 Model Performance Comparison Table

Model	Test RMSE	R ² Score	Rank
Linear Regression	~0.7255	~0.576	2nd
Ridge Regression	~0.7254	~0.576	1st (Best)
Decision Tree (max_depth=5)	~0.7200	~0.580	3rd

4.3 Visual Validation

Actual vs Predicted scatter plots were generated for all three models. A closer alignment of data points to the red reference line (perfect prediction line) indicates better model performance. Ridge Regression showed the tightest clustering around the reference line among the linear models.

5. Final Model Selection & Justification

Selected Model: Ridge Regression (alpha=1.0)

After comparing all three models on RMSE, R^2 Score, and generalization behavior, Ridge Regression was selected as the final model for the following reasons:

- Best overall balance of RMSE and R^2 Score on the test set.
- L2 regularization (alpha=1.0) prevents overfitting by penalizing large coefficients.
- Handles multicollinearity between features better than plain Linear Regression.
- Simple and interpretable — suitable for professional portfolios and production deployment.
- Scales well with the number of features without significant performance degradation.

6. Conclusion

Task 2 successfully demonstrated a complete, professional Machine Learning workflow. Starting from raw data loading through feature scaling, model training, and rigorous evaluation, the project mirrors real-world ML engineering practices. Key takeaways include:

- Feature scaling is essential — it directly impacts model performance and stability.
- Multiple models should always be compared before selecting a final solution.
- RMSE and R^2 together provide a comprehensive picture of model quality.
- Ridge Regression outperforms plain Linear Regression when multicollinearity is present.
- Visual validation (Actual vs Predicted plots) confirms numerical metrics intuitively.